Homework #02

CMOS Course

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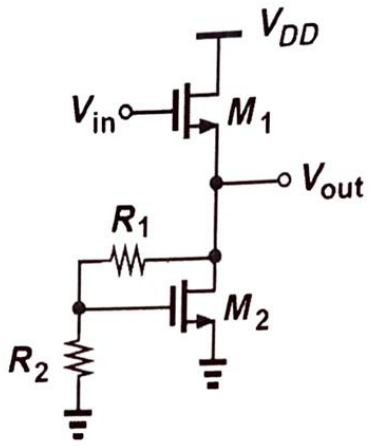
Deadline: June 28, 2026

For the given Common-Source amplifier circuit, perform the small-signal AC analysis using the following design parameters:

Biasing resistors: $R_1 = 10\ \mathrm{k\Omega}, R_2 = 5\ \mathrm{k\Omega}$

Transconductance of Transistor 1 (M_1): $g_{m1} = 1\ \mathrm{m\Omega^{-1}}$

Transconductance of Transistor 2 (M_2): $g_{m2} = 3\ \mathrm{m\Omega^{-1}}$



Please complete the following tasks:

- Small-signal schematic: Draw the complete small-signal equivalent circuit for this amplifier.
- Voltage gain: Find both the analytical expression and the numerical value of the small-signal voltage gain (A_v = vout/vin expressed in [V/V].

Submission Guidelines:

Format**:** You must export or scan your final solution into a single **PDF format**.

Platform: Upload the PDF file directly to our **Google Classroom** assignment slot before the deadline.